

CHEMISTRY 30

UNIT C STUDENT HANDBOOK

Chemical Changes of Organic Compounds

Learning Objectives

By the end of this unit, you will be able to:

- Analyze carbon bonding and geometry by linking hybridization (sp^3 , sp^2 , sp) to σ/π bonding, molecular shape, and reactivity trends.
- Derive and justify the distinction between saturated and unsaturated hydrocarbons using bond energy, electron density, and typical reaction pathways.
- Apply IUPAC nomenclature to: branched alkanes; alkenes/alkynes with locants; aromatic hydrocarbons; halogenated hydrocarbons; alcohols; carboxylic acids; and esters—using consistent parent-chain selection and lowest-locant reasoning.
- Construct and interpret condensed, expanded structural, and skeletal representations; translate between representations without changing connectivity.
- Systematically generate, classify, and justify constitutional isomers (chain/position/functional group), and predict how branching and functional groups affect physical properties.
- Predict physical-property trends (boiling point, solubility, volatility) by evaluating London dispersion, dipole–dipole forces, and hydrogen bonding in specific molecules.
- Predict reaction type and major organic product for key transformations: addition (X_2 , HX , hydration conceptually), substitution (haloalkanes under nucleophilic conditions), elimination (base-promoted formation of alkenes), combustion, esterification, and addition polymerization.
- Explain organic reaction outcomes mechanistically using bond-breaking/bond-forming logic (electron-rich π bond reactivity; leaving-group departure; base abstraction of β -hydrogen; condensation bond formation).
- Balance complete combustion equations and perform multi-step stoichiometry ($\text{mass} \leftrightarrow \text{moles} \leftrightarrow \text{mass}$), including unit tracking and significant-figure control.
- Determine empirical (and when sufficient information is provided, molecular) formula from combustion data; justify each conversion step.
- Evaluate polymer use and production from multiple STS lenses (environmental persistence, economic benefits, risk trade-offs, and mitigation strategies), grounding claims in chemical structure and properties.

Student Self-Verification Checklist

Use this checklist to confirm diploma-level readiness:

- I can define and correctly use: saturated/unsaturated, homologous series, functional group, σ bond, π bond, electrophile/nucleophile (concept level), substitution vs elimination, condensation vs addition polymerization.
- I can write chemical formulas using proper subscripts and ions using proper superscripts (e.g., Na^+ , SO_4^{2-} rendered with superscripts) using DOCX formatting—not plain text shortcuts.
- I can select the correct parent chain (including the chain that contains the multiple bond) and justify numbering using the lowest-locant rule.
- I can name and draw branched alkanes and alkenes with correct locants; I can reverse the process (name $\rightarrow$ structure) without changing connectivity.
- I can identify alcohol, carboxylic acid, ester, and haloalkane functional groups in unfamiliar structures.
- I can generate all constitutional isomers for a given formula systematically and prove I have no duplicates.
- I can predict boiling point differences by naming the dominant intermolecular force(s) and linking force strength $\rightarrow$ energy required $\rightarrow$ boiling point.
- I can predict solubility trends (polar vs nonpolar) by connecting functional group polarity to intermolecular attractions with water.
- Given reagents/conditions, I can predict whether addition, substitution, or elimination dominates and justify using structural and condition cues.
- I can explain (not just state) why alkenes undergo addition (π bond electron density; formation of stronger σ bonds).
- I can explain why elimination requires a base and a β -hydrogen and why alcoholic base conditions often favor elimination over substitution.
- I can explain esterification as a condensation process: identify which atoms leave as water and which fragments form the new ester linkage.
- I can describe addition polymerization as π -bond opening leading to a saturated chain, with the repeating unit written correctly.
- I can balance combustion equations before calculating; I can show a unit roadmap ($\text{g} \rightarrow \text{mol} \rightarrow \text{mol} \rightarrow \text{g}$) and track significant figures.
- I can determine empirical formula from combustion CO_2 and H_2O masses and justify each 1:1 or $\times 2$ relationship.
- I can interpret data tables (e.g., boiling points across a series) and explain trends using structure and intermolecular forces.
- I can identify common traps: choosing the wrong parent chain; numbering in the wrong direction; confusing coefficients with subscripts; assuming “heavier = higher bp” without checking hydrogen bonding; predicting substitution when conditions favor elimination.
- I can write responses in Step 1–5 format and include validation checks (trend check, unit check, reasonableness check).

Diploma-Level Practice Questions (with Fully Worked Solutions)

Question 1 | Complete Combustion Stoichiometry (Multi-step quantitative)

Student Task:

Complete Combustion Stoichiometry (Multi-step quantitative)

Propane undergoes complete combustion in excess oxygen.

Task: Calculate (a) the mass of CO_2 produced and (b) the mass of H_2O produced when 36.5 g of propane combusts.

Show balanced equation, unit tracking, and significant figures.

Step 1 — Identification

Given: mass of propane = 36.5 g; propane formula C_3H_8 . Need: mass of CO_2 and mass of H_2O produced.

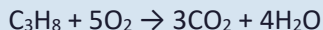
Relevant principle: Use coefficients in a balanced equation to convert moles of reactant to moles of products.

Step 2 — Principle Statement

In complete combustion, hydrocarbon + $\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$. After balancing, the mole ratios come directly from coefficients. Convert mass $\rightarrow$ moles, apply mole ratio, convert moles $\rightarrow$ mass.

Step 3 — Explicit Reasoning

1) Balance the combustion equation:



2) Convert mass of propane to moles:

Molar mass of propane, $M(\text{C}_3\text{H}_8) = 44.11 \text{ g}\cdot\text{mol}^{-1}$

$n(\text{propane}) = 36.5 \text{ g} \div 44.11 \text{ g}\cdot\text{mol}^{-1} = 0.827 \text{ mol}$ (3 sig figs).

3) Use mole ratios from the balanced equation: 1 mol propane $\rightarrow$ 3 mol CO_2 and 4 mol H_2O .

$n(\text{CO}_2) = 0.827 \times 3 = 2.48 \text{ mol}$.

$n(\text{H}_2\text{O}) = 0.827 \times 4 = 3.31 \text{ mol}$.

4) Convert moles to mass:

$m(\text{CO}_2) = 2.48 \text{ mol} \times 44.01 \text{ g}\cdot\text{mol}^{-1} = 109 \text{ g}$ (3 sig figs).

$$m(\text{H}_2\text{O}) = 3.31 \text{ mol} \times 18.02 \text{ g}\cdot\text{mol}^{-1} = 59.6 \text{ g (3 sig figs)}.$$

Pause and predict: Before calculating, you should expect total product mass > reactant hydrocarbon mass, because oxygen from O_2 is incorporated into products.

Step 4 — Validation

Unit check: $\text{g} \rightarrow \text{mol} \rightarrow \text{mol} \rightarrow \text{g}$ is consistent. Magnitude check: CO_2 mass substantially larger than propane mass is expected because each mole propane forms 3 moles CO_2 and oxygen adds mass.

Step 5 — Final Answer

$$m(\text{CO}_2) = 109 \text{ g}; m(\text{H}_2\text{O}) = 59.6 \text{ g}.$$

Common Error

Using an unbalanced equation or treating subscripts as coefficients (e.g., thinking CO_2 implies 2 mol CO). This mixes composition with stoichiometric ratio and produces wrong mole ratios.

Exam Strategy

Write the balanced equation first, then write a unit-roadmap line: $\text{g propane} \rightarrow \text{mol propane} \rightarrow \text{mol product} \rightarrow \text{g product}$. Don't start arithmetic until the roadmap is correct.

Question 2 | Combustion Analysis (Empirical AND Molecular Formula)

Student Task:

Combustion Analysis (Empirical AND Molecular Formula)

An unknown compound containing only carbon and hydrogen is combusted.

Products measured: 13.2 g CO_2 and 5.40 g H_2O .

(a) Determine the empirical formula.

(b) If the molar mass of the compound is $56.0 \text{ g}\cdot\text{mol}^{-1}$, determine the molecular formula.

Step 1 — Identification

Given: $m(\text{CO}_2)=13.2\text{ g}$, $m(\text{H}_2\text{O})=5.40\text{ g}$; compound is hydrocarbon (C and H only). Need: empirical formula and molecular formula given molar mass $56.0\text{ g}\cdot\text{mol}^{-1}$.

Relevant principle: In complete combustion, all C $\rightarrow$ CO_2 and all H $\rightarrow$ H_2O . Therefore $n(\text{C})=n(\text{CO}_2)$ and $n(\text{H})=2\cdot n(\text{H}_2\text{O})$.

Step 2 — Principle Statement

Convert measured CO_2 and H_2O masses to moles. Use 1:1 relationship for C from CO_2 and $\times 2$ relationship for H from H_2O . Reduce to simplest whole-number ratio for empirical formula. Use molar-mass ratio to scale to molecular formula.

Step 3 — Explicit Reasoning

- 1) Moles of CO_2 : $n(\text{CO}_2)=13.2 \div 44.01 = 0.300\text{ mol} \rightarrow n(\text{C})=0.300\text{ mol}$.
- 2) Moles of H_2O : $n(\text{H}_2\text{O})=5.40 \div 18.02 = 0.300\text{ mol} \rightarrow n(\text{H})=2\times 0.300 = 0.600\text{ mol}$.
- 3) Ratio C:H = $0.300 : 0.600 = 1 : 2 \rightarrow$ empirical formula = CH_2 .
- 4) Empirical formula mass: $M(\text{CH}_2) = (12.01) + 2(1.01)=14.03\text{ g}\cdot\text{mol}^{-1}$.
- 5) Molecular factor: $56.0 \div 14.03 \approx 3.99 \approx 4$.
- 6) Molecular formula = $(\text{CH}_2)_4 = \text{C}_4\text{H}_8$.

Pause and predict: If empirical formula is CH_2 , the compound likely matches the general formula C_nH_{2n} (alkene or cycloalkane).

Step 4 — Validation

Check scaling: $4\times 14.03 = 56.12\text{ g}\cdot\text{mol}^{-1}$, consistent with given $56.0\text{ g}\cdot\text{mol}^{-1}$ within rounding. Check ratio: C_4H_8 reduces to CH_2 .

Step 5 — Final Answer

Empirical formula: CH_2 . Molecular formula: C_4H_8 .

Common Error

Forgetting that each H_2O contains 2 H atoms, leading to $n(\text{H})=n(\text{H}_2\text{O})$ instead of $2\cdot n(\text{H}_2\text{O})$. This halves hydrogen and changes the formula.

Exam Strategy

Write two explicit extraction rules at the top: $n(\text{C})=n(\text{CO}_2)$; $n(\text{H})=2\cdot n(\text{H}_2\text{O})$. Then reduce ratio and only then use molar-mass scaling.

Question 3 | Compare-and-Contrast: Isomerism and Physical Properties (Extended response)

Student Task:

Compare-and-Contrast: Isomerism and Physical Properties (Extended response)

Consider the three constitutional isomers of C_5H_{12} : pentane, 2-methylbutane, and 2,2-dimethylpropane.

Task: (a) Explain why they are constitutional isomers.

(b) Predict the relative boiling points (highest $\rightarrow$ lowest) and justify using intermolecular forces and surface area.

(c) Identify one diploma trap in isomer questions and correct it.

Step 1 — Identification

Given: three compounds share molecular formula C_5H_{12} . Need: explain isomerism, predict bp order, justify using intermolecular forces (London dispersion) and molecular shape.

Relevant principle: Constitutional isomers have same molecular formula but different connectivity; boiling point for nonpolar alkanes depends largely on London dispersion strength, which increases with surface area (less branching $\rightarrow$ greater surface area $\rightarrow$ stronger dispersion $\rightarrow$ higher bp).

Step 2 — Principle Statement

For nonpolar molecules of the same formula, the dominant intermolecular force is London dispersion. More extended shapes have greater contact surface $\rightarrow$ stronger instantaneous dipole interactions $\rightarrow$ higher boiling point. Branching reduces surface area and lowers boiling point.

Step 3 — Explicit Reasoning

(a) Why constitutional isomers: They have the same count of atoms (C_5H_{12}) but different carbon skeleton connectivity (straight vs branched). The arrangement of bonds differs, so they are distinct substances with different properties.

(b) Boiling point order prediction:

- Pentane (straight chain) has the largest surface area $\rightarrow$ strongest London dispersion $\rightarrow$ highest bp.

- 2-methylbutane (one branch) has less surface area than pentane → intermediate bp.
- 2,2-dimethylpropane (highly branched, compact) has the smallest surface area → weakest dispersion → lowest bp.

Therefore: pentane > 2-methylbutane > 2,2-dimethylpropane (highest to lowest bp).

Pause and predict: If a student says 'more branching = higher bp because it's heavier,' that should trigger a correction: all three have identical molar mass, so mass cannot explain differences.

(c) Diploma trap + correction: Trap—treating different names as different isomers when they are the same structure renumbered (e.g., 3-methylbutane vs 2-methylbutane). Correction—apply lowest-locant numbering; if renumbering makes names identical, they are the same compound, not a new isomer.

Step 4 — Validation

Check logic: All are nonpolar; only dispersion differs. More branching → smaller surface area → lower bp is consistent with known alkane trends.

Step 5 — Final Answer

They are constitutional isomers because they share C_5H_{12} but differ in connectivity. Boiling points: pentane highest, then 2-methylbutane, then 2,2-dimethylpropane. Key trap: duplicate isomers via renumbering; fix by applying lowest-locant rule.

Common Error

Claiming the most branched isomer has the highest bp because it is 'more complex.' Complexity is not a force; surface area and intermolecular attractions are.

Exam Strategy

When predicting bp among isomers: (1) identify dominant IMF, (2) compare surface area/branching, (3) state trend with cause → effect, (4) provide a quick sanity check (same molar mass).

Question 4 | IUPAC Nomenclature and Structure Translation (Higher-order application)

Student Task:

IUPAC Nomenclature and Structure Translation (Higher-order application)

A compound has the condensed structure: $\text{CH}_3\text{--CH}(\text{CH}_3)\text{--CH}_2\text{--CH}(\text{CH}_2\text{CH}_3)\text{--CH}_3$.

Task: (a) Provide the correct IUPAC name with justification of parent chain and numbering.

(b) Write the condensed structure of the isomer that moves the ethyl substituent to the adjacent carbon while keeping the same parent chain length.

(c) Identify a common naming trap and how to avoid it.

Step 1 — Identification

Given a condensed structural formula with branching. Need IUPAC name and an isomer produced by changing substituent position. Principle: longest continuous chain as parent; number for lowest set of locants; identify substituents and alphabetize (ignoring di-, tri-).

Step 2 — Principle Statement

IUPAC naming: (1) find the longest chain, (2) number to give the lowest possible locants, (3) identify substituents with locants, (4) list substituents alphabetically, (5) write parent name with appropriate ending.

Step 3 — Explicit Reasoning

(a) Parent chain selection: The longest continuous chain contains 5 carbons (pentane). (Note: tracing through the ethyl branch does not create a 6-carbon continuous chain without breaking connectivity.)

Identify substituents on the pentane chain:

- At carbon 2: a methyl group (CH_3).
- At carbon 4: an ethyl group (CH_2CH_3).

Numbering decision: If numbered left→right, locants are 2-methyl and 4-ethyl. If numbered right→left, locants are 2-ethyl and 4-methyl. Compare lowest set: {2,4} vs {2,4} tie, so use alphabetical priority for the first point of difference: ethyl (E) before methyl (M), so ethyl should get the lower number in the tie case → choose numbering that gives ethyl at 2.

Therefore name: 2-ethyl-4-methylpentane.

(b) Positional isomer: Move ethyl from carbon 2 to adjacent carbon 3 while keeping pentane parent: 3-ethyl-4-methylpentane (condensed: $\text{CH}_3\text{--CH}_2\text{--CH}(\text{CH}_2\text{CH}_3)\text{--CH}(\text{CH}_3)\text{--CH}_3$).

(c) Trap: accidentally choosing a non-longest parent chain or numbering without checking tie-break rules. Avoid by explicitly writing both numbering options, listing the locant sets, then applying tie-breaker systematically.

Step 4 — Validation

Check parent chain length: 5 carbons. Check substituent identities and positions match the structure. Tie-breaker applied consistently.

Step 5 — Final Answer

IUPAC name: 2-ethyl-4-methylpentane. Positional isomer (ethyl moved): 3-ethyl-4-methylpentane with condensed structure $\text{CH}_3\text{--CH}_2\text{--CH}(\text{CH}_2\text{CH}_3)\text{--CH}(\text{CH}_3)\text{--CH}_3$.

Common Error

Numbering to make 'methyl' lowest automatically, forgetting tie-break rules or alphabetization.

Exam Strategy

Write two candidate numberings, list locant sets, then apply: lowest set $\rightarrow$ first point of difference $\rightarrow$ alphabetical tie-break if needed. Show this explicitly for full marks.

Question 5 | Mechanism-Based Prediction: Addition of HX to an Alkene (Mechanism)

Student Task:

Mechanism-Based Prediction: Addition of HX to an Alkene (Mechanism)

Propene reacts with hydrogen bromide (HBr) at room temperature (no peroxides).

Task: (a) Predict the major organic product.

(b) Explain mechanistically (bond logic) why that product is favored (Markovnikov reasoning without formal carbocation diagrams).

(c) Identify the diploma trap involving anti-Markovnikov conditions.

Step 1 — Identification

Given: propene (an alkene) + HBr, no peroxides. Need: major product and mechanistic justification.

Principle: In electrophilic addition to an alkene, the π bond breaks and forms two new σ bonds. The major product forms via the pathway that creates the more stable intermediate / more substituted arrangement (Markovnikov outcome).

Step 2 — Principle Statement

Addition of HX to an unsymmetrical alkene favors placing H on the carbon that leads to the more substituted (more stabilized) arrangement for the halogen-bearing carbon. Mechanistically: the electron-rich π bond attacks H (electrophile), and Br attaches such that the more stable intermediate pathway dominates.

Step 3 — Explicit Reasoning

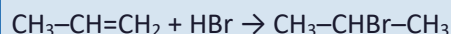
Propene has structure $\text{CH}_3\text{--CH=CH}_2$. Two possible addition orientations:

Path A: H adds to terminal carbon (CH_2), Br ends up on middle carbon $\rightarrow$ 2-bromopropane.

Path B: H adds to middle carbon, Br ends up on terminal carbon $\rightarrow$ 1-bromopropane.

Why A favored: placing Br on the more substituted carbon corresponds to the more stabilized pathway (secondary vs primary intermediate concept). Therefore major product is 2-bromopropane.

Representative equation:



Pause and predict: If a question mentions peroxides (ROOR) or radical conditions, your predicted major product may flip (anti-Markovnikov). Since peroxides are absent here, do NOT flip.

Step 4 — Validation

Check atoms conserved: reactants $\text{C}_3\text{H}_6 + \text{HBr} \rightarrow$ product $\text{C}_3\text{H}_7\text{Br}$. Product identity matches addition across the double bond (no double bond remains).

Step 5 — Final Answer

Major product: 2-bromopropane. Mechanistic justification: addition proceeds via pathway favoring the more substituted arrangement (Markovnikov).

Common Error

Automatically using anti-Markovnikov reasoning without checking for peroxide/radical conditions.

Exam Strategy

Always scan for condition cues first (peroxides, UV, heat). If none are present, default to Markovnikov for HX addition and justify by stability/ substitution reasoning.

Question 6 | Competing Pathways: Substitution vs Elimination (Higher-order application)

Student Task:

Competing Pathways: Substitution vs Elimination (Higher-order application)

Consider 2-bromobutane reacting with hydroxide ion.

Case 1: aqueous NaOH, warm.

Case 2: alcoholic KOH, hot (reflux).

Task: For each case, predict the major organic product and justify why (conditions → mechanism). Include the concept of β -hydrogen for elimination.

Step 1 — Identification

Given: haloalkane (2-bromobutane) with a leaving group (Br). Reagent OH^- can act as nucleophile (substitution) or base (elimination). Conditions determine dominant pathway.

Principle: Aqueous conditions favor substitution (formation of alcohol). Alcoholic, hot, strong base conditions favor elimination (formation of alkene). Elimination requires a β -hydrogen on a carbon adjacent to the carbon bearing the leaving group.

Step 2 — Principle Statement

Substitution replaces the leaving group with a nucleophile (OH^-) to form an alcohol. Elimination removes HX to form an alkene when a strong base abstracts a β -hydrogen while the leaving group departs. Temperature and solvent strongly influence which pathway dominates.

Step 3 — Explicit Reasoning

Case 1 (aqueous NaOH, warm): OH^- is well-solvated; nucleophilic substitution is favored. Major product: 2-butanol ($\text{CH}_3\text{--CH(OH)--CH}_2\text{--CH}_3$).

Case 2 (alcoholic KOH, hot): OH^-/RO^- behaves as a strong base; heat favors elimination (entropy). The base removes a β -hydrogen (on C_1 or C_3 relative to the $\text{C}_2\text{--Br}$ carbon), Br leaves, and a C=C forms.

Possible alkenes from elimination: 1-butene ($\text{CH}_2\text{=CH--CH}_2\text{--CH}_3$) and 2-butene ($\text{CH}_3\text{--CH=CH--CH}_3$). More substituted alkene (2-butene) is typically major (Zaitsev trend) in elimination.

Therefore: Case 2 major product: 2-butene (mixture of cis/trans possible; unless specified, state 2-butene as major).

Pause and predict: Students often forget that the same reagent OH^- can give different products depending on solvent and temperature. Conditions are part of the 'given information.'

Step 4 — Validation

Check functional group changes: substitution yields alcohol; elimination yields alkene. Both are chemically reasonable. β -hydrogens exist on adjacent carbons, so elimination is feasible.

Step 5 — Final Answer

Case 1: 2-butanol (substitution). Case 2: 2-butene as major (elimination).

Common Error

Predicting substitution in alcoholic, hot conditions because OH^- is recognized only as a nucleophile. This ignores base behavior and temperature effects.

Exam Strategy

Make a conditions decision tree: aqueous $\rightarrow$ substitution; alcoholic + heat $\rightarrow$ elimination. Then for elimination, identify β -hydrogens and predict the more substituted alkene as major.

Question 7 | Esterification and Equilibrium (Quantitative reasoning)

Student Task:

Esterification and Equilibrium (Quantitative reasoning)

Ethanoic acid reacts with ethanol to form ethyl ethanoate and water (equilibrium).

A reaction mixture starts with 0.500 mol ethanoic acid and 0.500 mol ethanol, and no products.

At equilibrium, 0.180 mol of ester is present.

Task: (a) Determine the equilibrium amounts of all species.

(b) Compute K_c for the reaction using concentrations if the mixture volume is 1.00 L.

(c) Explain why catalysts do not change K_c .

Step 1 — Identification

Given initial moles of acid and alcohol; equilibrium moles of ester. Need equilibrium moles of all species, then K_c using 1.00 L so $\text{mol} = \text{mol} \cdot \text{L}^{-1}$ numerically.

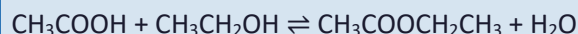
Principle: ICE table for equilibrium. Stoichiometry 1:1:1:1. $K_c = [\text{ester}][\text{water}]/([\text{acid}][\text{alcohol}])$.

Step 2 — Principle Statement

For a 1:1:1:1 equilibrium, if x moles of ester form, x moles of each reactant are consumed and x moles of water are produced. Concentrations at equilibrium determine K_c . Catalysts change rate, not equilibrium position, so K_c is unchanged at a given temperature.

Step 3 — Explicit Reasoning

Reaction equation (condensed):



Let x = moles of ester formed = 0.180 mol (given).

Initial: acid 0.500, ethanol 0.500, ester 0, water 0.

Change: acid $-x$, ethanol $-x$, ester $+x$, water $+x$.

Equilibrium: acid $0.500 - 0.180 = 0.320$ mol; ethanol 0.320 mol; ester 0.180 mol; water 0.180 mol.

Volume = 1.00 L $\rightarrow$ concentrations equal moles ($\text{mol} \cdot \text{L}^{-1}$).

$$K_c = (0.180)(0.180) / (0.320)(0.320) = 0.0324 / 0.1024 = 0.316 \text{ (3 sig figs)}.$$

Catalyst note: Catalyst lowers activation energy for forward and reverse equally, so equilibrium ratio (K_c) stays the same; only time to reach equilibrium changes.

Step 4 — Validation

Check: reactants decreased by the same amount as products increased (1:1 stoichiometry). All equilibrium amounts are nonnegative. K_c magnitude (<1) indicates reactants favored, consistent with esterification being equilibrium-limited.

Step 5 — Final Answer

Equilibrium moles: acid 0.320 mol; ethanol 0.320 mol; ester 0.180 mol; water 0.180 mol. $K_c = 0.316$. Catalysts do not change K_c at fixed temperature.

Common Error

Using 0.180 as the remaining reactant rather than the formed product, or forgetting that water is produced in equal amount to ester.

Exam Strategy

Write a 4-line ICE table. Label x clearly. Substitute x with the given equilibrium amount. Only after you compute all equilibrium values should you plug into K_c .

Question 8 | Addition Polymerization and Degree of Polymerization (Quantitative + mechanism)

Student Task:

Addition Polymerization and Degree of Polymerization (Quantitative + mechanism)

Ethene undergoes addition polymerization to form polyethylene.

Task: (a) Write the repeating unit correctly.

(b) A sample of polyethylene has an average molar mass of $84,000 \text{ g}\cdot\text{mol}^{-1}$. Estimate the average degree of polymerization n .

(c) Explain, using bond logic, what happens to the π bond during polymerization.

Step 1 — Identification

Given: polyethylene average molar mass $84,000 \text{ g}\cdot\text{mol}^{-1}$. Need repeating unit and n . Principle: repeating unit from ethene is $-\text{CH}_2-\text{CH}_2-$ with formula C_2H_4 (14? actually $28.05 \text{ g}\cdot\text{mol}^{-1}$). Degree of polymerization $n \approx M(\text{polymer})/M(\text{repeating unit})$. Mechanism: π bond opens to form new σ bonds along chain.

Step 2 — Principle Statement

In addition polymerization, the alkene π bond breaks and monomers link via new C–C σ bonds, producing a saturated chain with repeating unit (C_2H_4). The degree of polymerization is the number of repeating units per chain on average: $n = M_{\text{polymer}} / M_{\text{repeat}}$.

Step 3 — Explicit Reasoning

(a) Repeating unit: $[-\text{CH}_2-\text{CH}_2-]_n$.

(b) $M(\text{repeating unit C}_2\text{H}_4) = 2(12.01) + 4(1.01) = 24.02 + 4.04 = 28.06 \text{ g}\cdot\text{mol}^{-1}$.

$n = 84,000 \div 28.06 = 2993 \approx 3.00 \times 10^3$ (3 sig figs).

(c) Bond logic: Each ethene has one σ and one π bond between carbons. Polymerization breaks the π bond (weaker, electron-rich) and replaces it with two new σ bonds—one to the previous monomer and one to the next—creating a long σ -bonded backbone.

Pause and predict: If you still draw double bonds in polyethylene, you have not applied the π -bond opening idea; that's a common diploma trap.

Step 4 — Validation

Reasonableness: n on the order of 10^3 is typical for many polymers. Units: $(\text{g}\cdot\text{mol}^{-1})/(\text{g}\cdot\text{mol}^{-1}) = \text{dimensionless}$, as required for n .

Step 5 — Final Answer

Repeating unit: $[-\text{CH}_2-\text{CH}_2-]_n$. Degree of polymerization $n \approx 3.00 \times 10^3$. π bond breaks and is replaced by σ bonds linking monomers.

Common Error

Using molar mass of ethene ($28 \text{ g}\cdot\text{mol}^{-1}$) but then forgetting polymer end groups are negligible for large n , or leaving double bonds in the repeating unit.

Exam Strategy

Write the repeating unit first, compute its molar mass, then do $n = M_{\text{polymer}} / M_{\text{repeat}}$ and sanity-check magnitude (10^2 – 10^5 depending on polymer).

Question 9 | Data Analysis: Boiling Point Trends (Data interpretation + proportional reasoning)

Student Task:

Data Analysis: Boiling Point Trends (Data interpretation + proportional reasoning)

Boiling points ($^{\circ}\text{C}$) for a set of compounds are shown:

- Ethane: -89

- Propane: -42
- Butane: -1
- Ethanol: 78

Task: (a) Explain the trend across the alkane series (ethane $\rightarrow$ butane).

(b) Explain why ethanol does not follow the alkane trend despite having a similar number of carbons to propane.

(c) Identify what intermolecular force dominates ethanol's boiling point.

Step 1 — Identification

Given boiling points for small alkanes and ethanol. Need trend explanation and force identification. Principle: For alkanes, London dispersion increases with molar mass and surface area; ethanol has hydrogen bonding, dramatically increasing bp.

Step 2 — Principle Statement

Within a homologous series where the dominant IMF is the same, boiling point increases with increasing dispersion forces (larger electron cloud, greater polarizability, larger surface area). Introducing a new stronger IMF (hydrogen bonding) can override this trend.

Step 3 — Explicit Reasoning

(a) Ethane $\rightarrow$ propane $\rightarrow$ butane: As carbon chain length increases, molar mass and surface area increase, strengthening London dispersion forces. Stronger attractions require more energy to separate molecules, so bp rises ($-89 \rightarrow -42 \rightarrow -1$ °C).

(b) Ethanol vs propane: Even though ethanol has only two carbons, it has an -OH functional group that enables hydrogen bonding. Hydrogen bonding is much stronger than dispersion forces in small alkanes, so ethanol's bp is far higher (78 °C).

(c) Dominant IMF in ethanol: hydrogen bonding between O-H and O lone pairs (in addition to dipole-dipole and dispersion).

Pause and predict: If you answer 'ethanol higher because heavier,' you will lose marks—ethanol is not heavier than butane, yet it has a higher bp due to IMF type.

Step 4 — Validation

Trend check: Increasing alkane size increases bp—consistent with known physical properties. Ethanol's large deviation aligns with hydrogen bonding effects.

Step 5 — Final Answer

Alkane bp increases with chain length due to stronger dispersion forces. Ethanol's high bp is due to hydrogen bonding from the -OH group, which dominates its boiling behavior.

Common Error

Treating boiling point as a function of 'number of carbons only' and ignoring functional groups.

Exam Strategy

For data trends: state the IMF type first, then connect structural change $\rightarrow$ IMF strength change $\rightarrow$ bp change. If a functional group appears, check for hydrogen bonding immediately.

Question 10 | STS Evaluation: Polyethylene Production and Use (Extended STS response)

Student Task:

STS Evaluation: Polyethylene Production and Use (Extended STS response)

Polyethylene (a common packaging plastic) is produced from ethene.

Task: Write a structured STS evaluation that includes:

- Chemical basis: why polyethylene is durable and persistent
- At least two benefits (social/economic)
- At least two risks (environmental/ethical)
- Mitigation strategies (chemistry/engineering/policy)
- A justified conclusion that weighs trade-offs (not one-sided).

Step 1 — Identification

Topic: Polyethylene lifecycle impacts. Need a multi-perspective evaluation grounded in chemical structure and properties.

Principle: STS responses earn marks by connecting chemistry (structure → properties → behavior) to societal outcomes, presenting benefits and risks, and concluding with a justified recommendation.

Step 2 — Principle Statement

Polyethylene is a nonpolar, saturated hydrocarbon polymer with strong C–C and C–H bonds. Chemical inertness and low polarity reduce biodegradation and increase persistence. These same features provide durability and usefulness. STS evaluation must weigh benefits against long-term environmental costs and propose realistic mitigation.

Step 3 — Explicit Reasoning

Chemical basis (cause → effect):

- Polyethylene's backbone is repeating $\text{--CH}_2\text{--CH}_2\text{--}$ with strong σ bonds. Microbes and water have limited chemical pathways to break these bonds → slow degradation.
- Nonpolarity and hydrophobicity reduce interaction with water and many reactive species → persistence in environments.

Benefits (examples):

- Social: Lightweight packaging reduces food spoilage and contamination; medical and safety applications (sterile packaging, protective materials).
- Economic: Low cost per unit mass; reduced transportation emissions due to low weight; large industrial employment base.

Risks (examples):

- Environmental: Persistence leads to accumulation in landfills and oceans; fragmentation produces microplastics that enter food webs; wildlife ingestion and habitat impacts.
- Ethical/Equity: Waste burden often falls on communities near landfills/incinerators; long-term environmental costs are shifted to future generations.

Mitigation strategies (linked to chemistry/engineering/policy):

- Design for reuse: prioritize durable reusable polyethylene where lifecycle emissions drop relative to repeated single-use alternatives.
- Improve recycling: better sorting, contamination reduction, and mechanical recycling where material quality supports it; consider chemical recycling where energy/emission profile is justified.
- Policy tools: extended producer responsibility, deposit systems, labeling standards, and targeted bans/fees for unnecessary single-use products.

Justified conclusion:

Polyethylene remains justified for applications where its durability directly reduces harm (medical, long-life goods, essential packaging), but widespread single-use deployment is chemically misaligned with persistence. Responsible use requires reduction of unnecessary single-use items, strong recovery systems, and accountability across the supply chain.

Pause and predict: A one-sided response (“plastics are bad”) will not earn full marks; diploma marking rewards balanced trade-off reasoning anchored in chemistry.

Step 4 — Validation

Check completeness: Includes chemistry basis, benefits, risks, mitigation, and a weighed conclusion. Claims are linked to polymer structure and bonding.

Step 5 — Final Answer

A balanced STS conclusion: Polyethylene provides major benefits due to durability and low cost, but chemical persistence creates long-term environmental risk; therefore, reduce unnecessary single-use, strengthen recovery/recycling, and reserve use for high-benefit applications with lifecycle justification.

Common Error

Writing an opinion without chemistry linkage (no mention of C–C/C–H bonds, nonpolarity, or persistence mechanism), or listing only pros/cons without a weighed conclusion.

Exam Strategy

Start with chemistry (structure → property). Then do a 2×2: benefits (social/economic) and risks (environmental/ethical). Add mitigation (practical). End with a conditional recommendation that shows trade-off weighing.

PERFORMANCE BENCHMARK

If you can independently name and draw unfamiliar organic structures, generate and justify isomers, predict reaction outcomes from conditions and bonding, complete combustion/empirical-formula calculations with unit tracking, interpret data trends using intermolecular forces, and write a balanced STS evaluation grounded in chemistry, you are operating at a 95%+ diploma level.

FINAL BLUE CLOSURE

Unit C mastery is one chain of reasoning: structure → bonding → functional group → mechanism → products → evidence → impacts. On diploma questions, you earn marks by showing that chain explicitly, step-by-step, with correct notation and validation.